Conceptual & Design Document - Airline Ground Handling Cost Management Platform

We envisage building a platform that could be used by airline ground handling service providers to invoice airlines in industry standard IATA formats (Level - 1) . Once the platform has a critical mass of suppliers, the same could be used by airlines to get a market view of the cost prevalent in different airports for different services (Level - 2) . If the airline is cooperative, this could be morphed into a full fledged invoicing & dispute management system (Level - 3) .

Business Need / Gap

1. Ground handling industry has only a few large scale players. Most of the suppliers provide services only on a regional basis for a few airports or for a few services. These providers lack sophisticated invoicing systems by and large . Most of the international airlines have specialized invoice processing systems that are geared up to handle industry standard IATA XML standards. By and large very few ground handlers provide invoices in IATA XML format. They invoice the airlines in PDF / paper
2. All the large airlines spend a lot of time converting these PDF invoices into electronic readable formats and verifying these invoices. Consequently, the payments to these ground handlers are largely delayed. If the invoices are in IATA XML formats, the processing of these for the airlines is straight through. It is a win-win situation for both airlines & ground handlers . For airlines , there is less human cost in processing these invoices. For ground handlers, the timely payment would be issued - resulting in better cash-flows
3. While the same XML standards are applicable for all the operating cost types of airlines (Fuel , Landing & Parking, Overflying , Hotels) , ground handling is what you can classify as a 'long tail' industry. The market share of the dominant players in industry is very less compared to fuel. The other cost types have the following challenges
    a. Overflying (civil aviation authorities are the service providers, which are the vendors) . Large number of authorities especially in Europe are IATA XML compliant.
    b. Landing & Parking (airports are the providers - but the invoicing is dependent on landing and parking timings - which are only available in their internal systems). Few airports of late (LHR , MUC , FRA) have adopted the IATA XML standards with encouraging results
    c. Hotels (This is another possible segment - But airlines are not the 'bread and butter' of hotels unlike ground handlers)

4. Because of the small nature of ground handlers, and their lack of invoicing systems, they too lack proper analytics on the cost base. The following questions typically puzzle them a lot
    a. What is the appropriate price to charge at a particular airport for a type of service (baggage handling / checkin counters etc) ?
    b. Is my pricing at a premium level or discount level ?
    c. What is my market share in a region or airport for a type of service ?
    d. What exactly is my revenue base service-wise ?

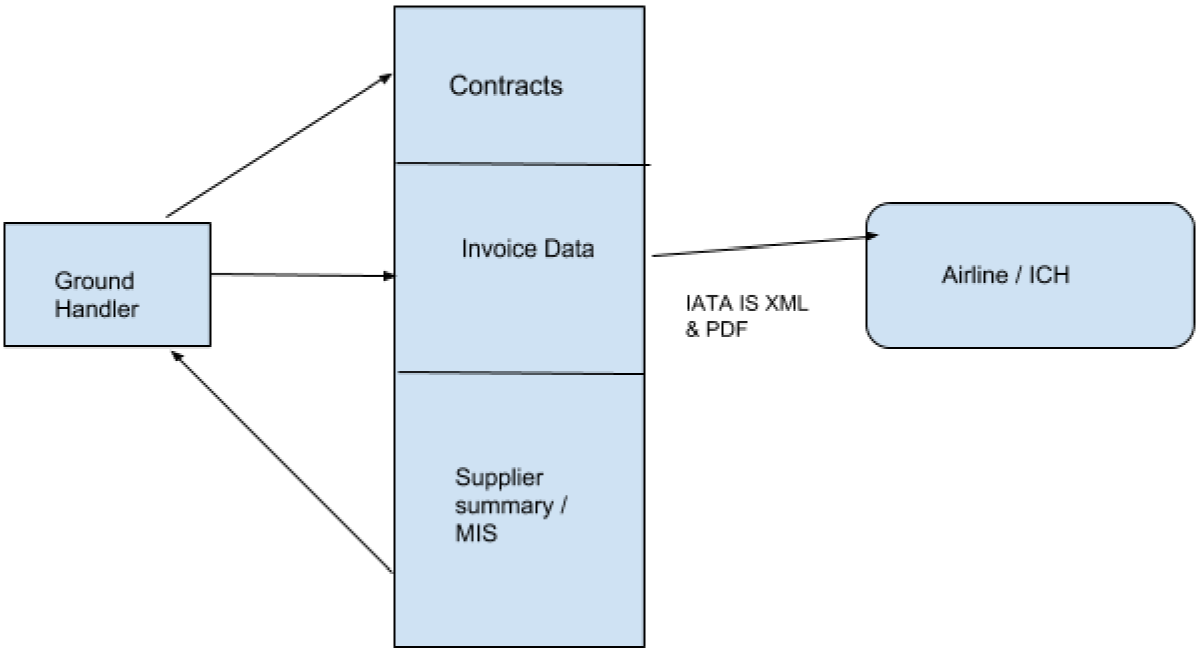
    Addressing many of these questions require having a critical mass of ground handlers and airlines in a region using this platform. This also would need us to carefully decide what data can we showcase. However, these are not immediate concerns
5. IATA Clearing House (ICH) - This is a mechanism by IATA in which you can mutually interact with other airlines & suppliers in a clearing-house manner. All international airlines are by default part of that (because of interline billing and settlement) . This is a great place for suppliers to get into if you are in a 'net-receivable' situation. When you invoice through ICH, airlines are bound to issue you immediate settlement without waiting for clearing issues / resolving issues etc. One amongst the many conditions for a ground handler to enter this is invoicing in IATA XML . Very few ground handlers are part of this
6. For airlines also, accurately estimating ground handling costs is a big challenge. When an airline is evaluating flights to a new airport, estimating the ground handling cost base in that airport is very difficult. The following questions puzzle airlines a lot
    a. What is the ground handling cost going to be in an airport ?
    b. Am I paying a premium price or discounted price at an airport ?
    c. Who are the potential service providers in a new airport ?
7. Dispute Management. Today, all disputes between an airline & ground handler is done over email / phone etc. This results in no audit trail / easy tracking of claims etc. The platform , on successful participation from airline as well as a number of ground handlers can be used to manage disputes in a structured way

Business Design of the Platform

(Level - 1)

In this mode (which is the base level needed for go- live) , the platform is driven entirely by the ground handler. Ground handler do three things mainly in this platform

(i) Enter & Maintain Contracts (ii) Enter Invoice Data (iii) Send Invoice to the airline in IS XML / PDF (iv) Mark which one of the invoice is settled (tick box) (iv) Consume the limited MIS - which is supplier specific information at this stage

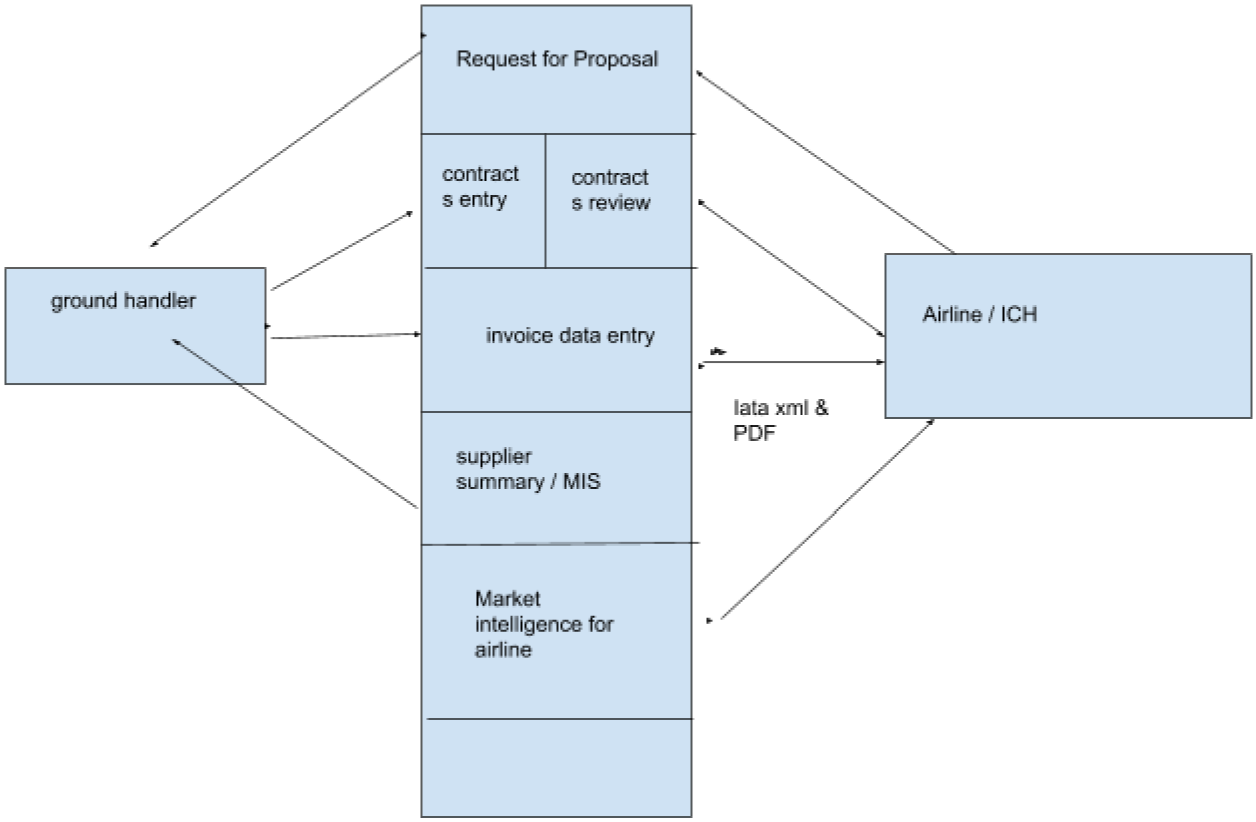


The main challenges in getting into this stage

(i) We need to begin from zero
(ii) Convincing small ground handlers, pilot with them and make the product perfect
(iii) If big suppliers become interested, GDS (flight data) integration

(Level - 2)

In this mode, the airline also participates in the platform in the form of reviewing & approving contracts, requesting ground handlers for a proposal. Airlines can see the market intelligence in an airport / region provided a sufficient number of suppliers become part of this. An airline can mark invoices as paid from their side in this (this will add more meaning to the MIS). An interface from various payment channels to the platform to get the paid status is too complex & diverse . So, not thinking about that now



In this level, there is more engagement from the airline. The idea is to drive this as a genuine ‘two-sided platform’ between airline & ground handler. Airline can initiate an RFP for a service in an airport , request review of a service contract to the ground handler. Before getting into this stage, the ground handler should be fully on-board with this system

Challenges

(i) For an airline to get into this platform for active collaboration with the ground handler, there should be a sizeable number of ground handlers who are using this platform. An airline won’t follow a process for a few ground handlers

(ii) Airlines are typically large organizations & slow to drive process change initiatives.

Level - 3

Level - 2 above plus dispute management . Airline will review the invoice data in the platform, for charges they see as incorrect , they will ‘ mark for dispute’ with appropriate reason , comments & justifying materials as attachments, raise a dispute. Supplier can put it back to the airline with justification & materials, Airlines can put it back to them. On successful ‘dispute acceptance’ by a ground handler, they will raise a ‘credit note’ (system should automatically facilitate that based on the accepted disputes) . The credit note will offset the open disputes that are accepted (again IATA XML format)

Challenges

This requires good close coordination & working between the airline & ground handler. Only one player cannot drive this. Before even attempting this, the critical mass supplier - airline combinations should be in Level - 2

At this level, we definitely need GDS integration. As a sizeable number of airlines - ground handlers come into this this level, we cannot afford to have an entirely manual keying in of flight details throughout.

IATA, ICH, IATA XML

IATA or International Air Transport Association is a confederation of Airlines & Airline related businesses. This is not a legal - body per se. But airlines use IATA as a front face to lobby governments / other industries etc for their benefits. IATA issues standards & guidelines to make mutual interaction & business in aviation industry smooth

IATA has engaged airlines in many aspects for a long time. The revenue side of international airlines (Fares , tickets , proration , interline , codeshare etc) is heavily driven by the IATA standards. However, the cost-side is not that much influenced by IATA. IATA XML / IS-XML / is a major attempt by IATA to standardize the cost management functions of aviation industry.

IATA IS XML is a standard that IATA has setup (not enforced as it cannot legally do so) to make mutual financial interaction (inter airline billing / airline - supplier transactions etc) smooth.

IATA clearing house(ICH) is a netting mechanism (A owes B $100 , B owes C $80 , C owes A $10 ⇒ Can be netted off to A paying C $70) with all interacting airlines & service providers. It is not easy for a small scale ground handler to get into this. It has lots of criteria (one of it is supporting IATA XML) for membership. The important point about clearing house is that, once an invoice is submitted into the clearing house, the participant is supposed to get the settlement as per the due date. An airline cannot hold the payment citing a dispute. Disputes are to be raised separately. The principle is ‘pay-first, dispute later’ . Of-course, if the volume of disputes is too large, IATA may very well kick out the participant from it.

The IATA IS XML standard is available here
https://www.iata.org/services/finance/financial/Documents/IS-XML_e-invoicing_standard_Ground_Handlers.pdf

This is yet to be assessed fully from a technical perspective. So, at this point we are unsure of any technical challenges for this XML generation / transmission. All the standards for this is open source. And changes keep coming in this driven by IATA.

There are similar XML standards for Fuel / Landing & Parking / Overflying etc

IATA XML groups different ground handling services into a finite set of charge codes given below.

Baggage, Baggage Delivery,Cargo Handling, Catering,Cleaning, Commission, Crew Accomodation, Crew Transportation,Customs Service Charge,Deicing,Departure Stamps,Immigration Fines, Lounges, Misc , Mishandling baggage, Mishandling Passenger,Motor Fuel, Passenger Handling, Passenger Transportation,Passenger Security,Ramp Handling,Rent Equipment,Stand, STPC,Utilities

So, when you define a contract, you are basically defining a service which maps into one of the charge codes above. You can name the service as anything. But it should be mapped into one of these charge codes so that XML generation is possible

Ground Handling Contracts

Ground Handling Contracts could be one among the below types

1. Unit Rate Based (Rate * Qty)
    a. This quantity could be anything (ramp, aerobridge , baggage handlers, checkin counters, de-icing fluid, baggage containers , blankets for passengers,…..) .. This is a big list. And it cannot be fully defined. Consequently from a design perspective, this has to be a user defined field
    b. Quantity could even be a product of two (checkin counters * hours of operation)
2. Slab based Type - 1
    a. Upto 3 wheelchairs in a flight $3, 3 - 5 Wheelchairs $5 etc
3. Slab based type - 2
    a. Upto 3 wheelchairs in a flight 0, if you choose 4th wheelchair entire qty (4* rate)
4. Time based rate
    a. Till 5 PM $100 per checkin counter per hour. 5PM - 10 PM $150 etc
5. Day / month based rates (Aerobridge charges are higher in the weekend)
6. Certain rates (basic handling charges for eg) are dependent on the MTOW (Maximum TakeOff weight) of an aircraft. This is specific for an airline. For eg, the MTOW of an A380 of SQ (Singapore Airlines) is different from MTOW of A380 of Etihad (EY). Certain ground handlers charge it by taking an industry reference MTOW rate. We may need to think about how to capture it . Every aircraft has a unique 'Tail ID' (Registration

number equivalent). We may need to maintain a self-updating list of tail number, reference industry weights, actual MTOW etc

Supplier MIS

The idea of detailing the supplier MIS is that, while we design this, we should consider the data requirements & architect in such a way that deriving these MIS is easy from a data & reporting perspective. The MIS is not an exhaustive one.This list is rather indicative

Supplier Financial Summary

Report Identifier	SFR1 (Will reference this in wireframes)
Report Description	Supplier Receivables Summary - This gives the summary of outstanding receivables of a supplier from different airlines & airport
Visualizations	Pie Chart giving airport-wise & airline-wise share of outstanding receivables for a ground handler. Dimensions - airline, airport , country , regional classification of airports & airlines specific to a supplier, currency , aircraft type Aging of receivables (sysdate - invoice date) ⇒ also analyzed by the above dimensions Another visualization would be a trend of absolute amount of receivables trending over the last few months (might be difficult as this would mean storing ‘as-of date’ data)
Drill down	To the actual numerical data of invoices
Level at which this would be a value add	Level - 1 itself

Report Identifier	SFR2
Report Description	Invoiced Amount airport & airline-wise
Visualizations	Bar graph of invoiced amount trending month-wise (Dimensions - airport , airline , regional classification , supplier specific classification of airline , aircraft type, currencies
Drill down	Drill down to the invoices
Level	**Level - 1**

Report Identifier	SFR3
Report Description	Revenue per invoiced Flight (airline -wise & airport-wise & service type-wise)
Visualizations	Line graph (dimensions - aircraft , airport , airline , regional classification , supplier specific classification of airline)
Drill Down	Nil
Level	Level - 1

Report Identifier	SFR4
Report Description	Amount waiting to be invoiced
Visualizations	Pie-chart (Airport , airline , regional classification , supplier specific classification of airline). Based on billing frequency , operational data (GDS) & contracts - we can calculate amounts that are yet to be invoiced
Drill down	To be invoiced amounts - airline , aircraft, service type wise etc
Level	Level - 2 / 3 (May need GDS interface)

Supplier Operations Summary

Report Identifier	SOR1
Report Description	Contracts up for Expiry
Visualizations	Tabular (Airline, Airport, Service Type , days before expiry) . Visualization with graphs will be more relevant once big value suppliers use this.
Drill Down	Nil
Level	Level -1

Report Identifier	SOR2
Report Description	Operational Footprint
Visualization	Map with dropdowns as Airline, Service Type
Drill Down	Contract Summary & SFR2
Level	Level - 2 / 3 (makes sense only for global players)

Report Identifier	SOR3
Report Description	Review & Request for Proposals . This gives the summary of contract review requests & request for proposal of a service at xyz airport from an airline
Visualization	Tabular. Map view when you reach level - 3
Drill downs	Proposal Details (tabular)
Level	Level - 2 / 3

Dispute Summary

Report Identifier	SDR1
Report Description	A bit too early to think about this. This report should give a summary of disputes raised by airline (4/5 types of disputes - operational data mismatch , contract rate/formula mismatch, Exchange Rate mismatch , referenced flight does not belong to us, misc)
VIsualization	Can be a pie chart based on the value of disputed line item. Tabular view is also essential
Drill Downs	Disputed line item & invoice details
Level	Level - 3

Airline MIS

This section details different reports / visualization that are relevant for an airline. Applicable only from Level -2 & 3

Airline Financial Summary

Report Identifier	AFR1
Report Description	Amounts billed by so and so suppliers, airport-wise , service-wise. Can show unpaid & total
Visualization	Pie-chart & bar-graph.
Drill down	Drill down into billed invoice details
Level	Level - 2

Report Identifier	AFR2
Report Description	Expected Billing Amounts (based on billing frequency)
Visualization	Line Graph (Day vs Amounts) - Airport-wise , service-wise & supplier-wise, currency-wise

Drill-down	Drill down into expected amounts
Level	Level - 2 /3

Airline Operations Summary

Report Identifier	AOR1
Report Description	Contracts Up for Expiry in so many days
Visualization	Tabular. Maybe a map-based airport-wise
Drill Down	Nil
Level	Level 2

Report Identifier	AOR2
Report Description	Current Footprint (Airports & Services)
Visualization	Map. On hovering in the map on an airport, services & suppliers could be highlighted with monthly rates & invoiced values appearing on mouseover.
Drill down	Contracts , invoice summary
Level	Level 2

Report Identifier	AOR3
Report Description	Request for Review summary
Visualization	Tabular. Airport-wise,service-wise & supplier-wise requests that has come for review of contracts
Drill Down	Nil
Level	Level - 2

Airline Dispute Summary

Report Identifier	ADR1
Report DEscription	Dispute summary . Similar to SDR1 , but from an airline perspective
Visualization	Tabular. Map-view / pie-view when the operations grow
Drill-down	To the disputed line items
Level	Level-3

Airport Cost Index

As it was detailed earlier, one of the most perplexing questions that route planners often encounter is, what is the handling cost going to be in so and so airport. There is an element of confidentiality here as suppliers won't like all of their data to be published outside. However, we can very well publish an aggregated index. Assuming there are 2 / 3 service providers in an airport who uses our platform, publishing an aggregate rate at an airport / region level is not a breach of confidentiality ideally

Cost per service type - dimensions - aircraft type , international , domestic etc

Airport service Provider's market

This is supplier's gateway to market their offering to an airline. On giving this view to the airline, if a supplier uses our services at an airport, those services will be listed as services available. Airlines can initiate request for proposal from this

This can also show to an airline, whether their existing rates are at a premium / discount when considering other suppliers in the region . For eg, calculate the unit rate of all services for all airlines, just publish whether this particular airlines unit rate is in the top-25% / Mid 50 % / bottom 25% (dimension - aircraft type)

(We need to debate a lot on the confidentiality aspect before we chose to do this - However these things are largely applicable at Level - 2 & 3)

Roles & Responsibilities

All the distinct Roles & Responsibilities that the platform should support is provided here. One user can have two or more roles

Ground Handler Roles

(service type/charge code as a restrictive dimension is not mandatory. More of a wish list)

Role Name	Description	Restrictive Dimensions
Contract Entry	One who enters the contracts	Airports , Airlines , Service Type
Admin	One who sets supplier-specific parameters, assigns user specific roles etc	
Contract Review & Approve	One who reviews & approves the entered contracts	Airport, Airlines,Service Types
Invoice Entry	One who keys in operational data for invoices	Airport, Airlines & Service types
Invoice Approver	One who approves & submits the invoice for sending to airline	Airport , Airline , Service Types
MIS view	Access to MIS views	Airport , Airline, Service Types
RFP monitoring & Response	Ability to monitor & respond to RFPs from airlines	Airport, Airline & service Type
Invoice Status Updation	Updating whether an invoice is paid or not	Airport, Airline & service Type
Dispute Handler	Ability to accept & respond to dispute, ability to request a credit note	Airport, Airline, Service TYpe
Dispute approver	One who authorizes credit notes for airlines	Airport, Airline, Service Type

Airline Roles

Role	Description	Restrictive Dimensions
INvoice Review	Ability to view invoices	Airport , Service Type
Invoice Disputer	Ability to raise dispute & close it	Airport , Service Type
Contract View	Ability to view airline specific contracts.	Airport, Service TYpe
Contract Review	Ability to raise a review request,	Airport, Service TYpe
RFP raise	ability to raise RFP	Airport, Service Type
MIS	Ability to see various MIS dashboards	Airport, Service Type
Update Paid status	Can update a paid status for an invoice	Airport, Service Type

Note: We may need to restrict access of various reports in the previous sections to specific airlines / suppliers . Many of the reports make sense only if you have enough data. So a report level access restriction specific to a customer of the platform is desirable. We may chose to do a pricing based on that as well

User - Stories

Ground Handler Contracts Entry

1. User clicks New Contract
2. Picks an Airline & Airport (Enabling an airline & airport for a ground handler is our job - depends on payment)
3. Select a service Type (One among the IATA Charge codes) & billing frequency (optional)
4. Enter From date - to Date
5. Enter a service name (supplier specific name)
6. Pick a currency
7. Pick a formula type (Refer the contracts section for different types of formulas)

8. The screen should render based on the formula chosen
9. Enter the rates, formula details if applicable , quantity driver & unit of measure (s) - The quantity driver (wheelchairs, check-in counters , deicing liquid) , will be visible when they key in the operational data for invoicing
10. Tax codes / rates applicable (again this could be tricky - need to study the iata xml properly to understand how that works
11. Save , Review with summary pops up
12. Submit

Ground Handler Contracts Approve

1. Contracts summary screen (airport , airline-wise)
2. page filters - waiting for approvals, approved, expired
3. Click view summary (should popup a summary of the contract with an approve button)
4. On Approval, the contract is finalized
5. Another button 'mark for review' with comments field - which would prompt the contract entering person to review it

Invoice Entry

1. Create New Invoice (Pick Airline & Airport)
2. Pick service(s) - An invoice can have multiple services . Each service belongs to a contract
3. On clicking next, system should prompt the user to enter the invoice header

Header Fields are as follows

(a) Invoice Date (Ideally this should be the system date. But most suppliers enter this a few days before. This should ideally be controlled by a supplier specific parameter.. Can become important if this reaches Level - 2 / 3 with active participation of airlines. Airlines simply hate supplier pre-dating the invoices. Their payment terms are based on invoice date. Hence if a supplier pre-dates it, it is problematic for airlines from a cash flow perspective

(b) Invoice Number (user enterable unique name - alphanumeric- per airline ground handler)

(c) Invoice currency. (Invoicing should ideally be in the contract currency. However in a few situations, supplier may enter invoices in a different currency. In such cases, an exchange rate field should become mandatory … When we reach level-2/ 3 we may need to build the capability of taking a feed of bloomberg/reuters exchange rate into the platform

(d) Invoice header total (can be set to auto-calculate as a total based on the lines)

(e) Invoice Due Date (later on we can try to build a payment terms logic - but at this point, a keyed in date field will do)

(f) Invoice From Date & To date (flight dates)
(g) Any other fields that may become mandatory after studying the IATA XML standard

For every service picked up, A set of lines should open up which allows user to key in the flight details which include

(a) Date (b) Flight Number © Aircraft registration (d) Dep (e) Destination (f) quantity drivers

- some of this will be mandatory , some are not.. Need to study the xml properly to understand those
- On entering the quantity drivers, system should auto calculate & populate the cost at a line level
- If a GDS integration is present (Level 2 / 3) , on entering the date ranges & some other parameter, system should automatically query & display the flights - which can be changed by user later

Save - Should validate & display the summary . Finalize - invoice entering person has submitted it

An invoice which is sent to supplier cannot be edited / changed. Till that point users should be able to change it.

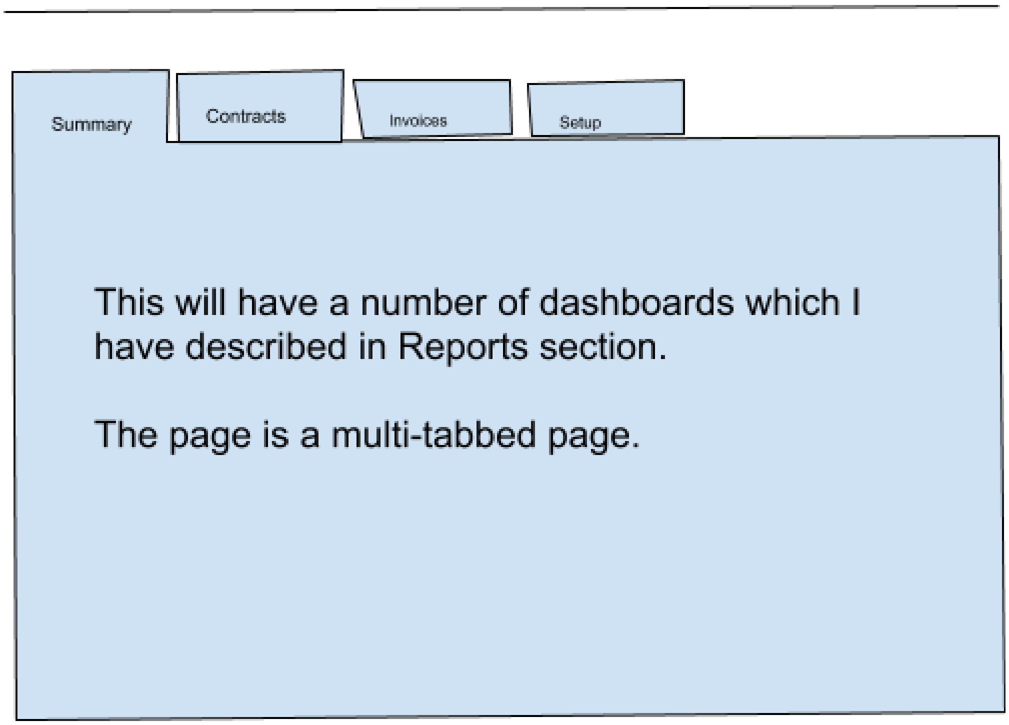
From approver role,

User should be able to review it..

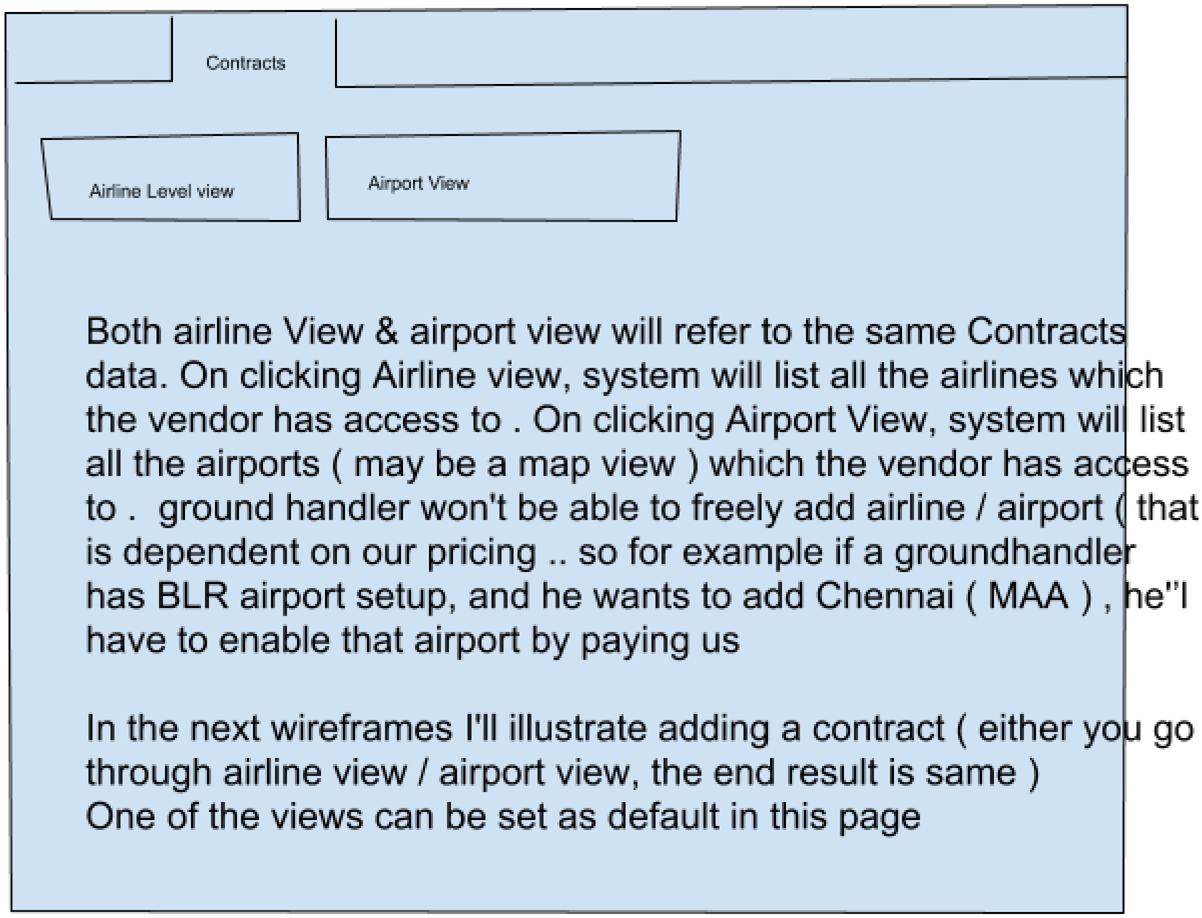
(1) Mark for modification with comments
(2) Approve it
(3) Send it to the relevant mail ID's (mail ID should be part of a ground handler - supplier specific setup)
(4) Mark a sent invoice as Paid
(5) Mark a sent invoice as disputed (In Level 3 you could have a more sophisticated mechanism) with comments

Wireframes (All at Level - 1)

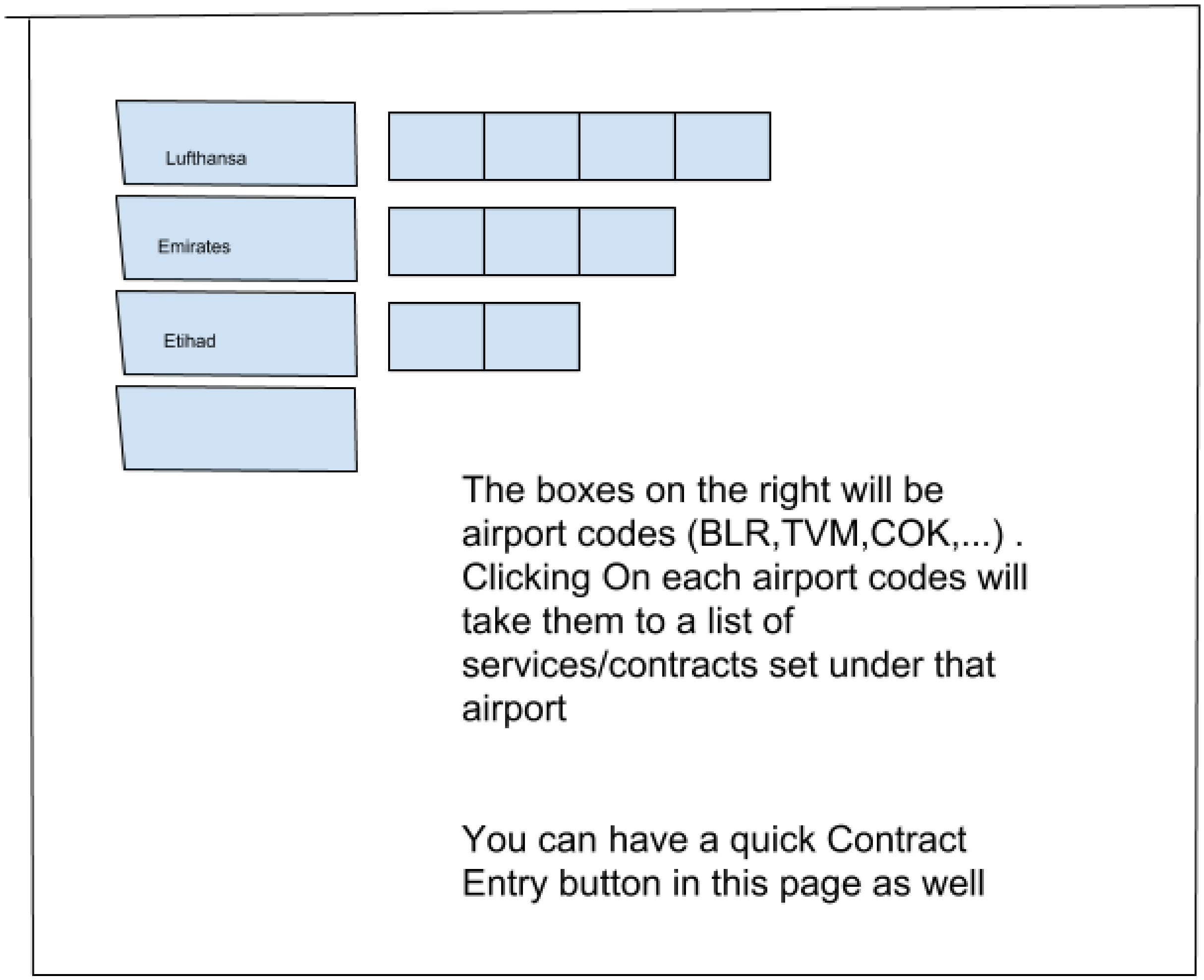
1) Landing Page for Ground Handler



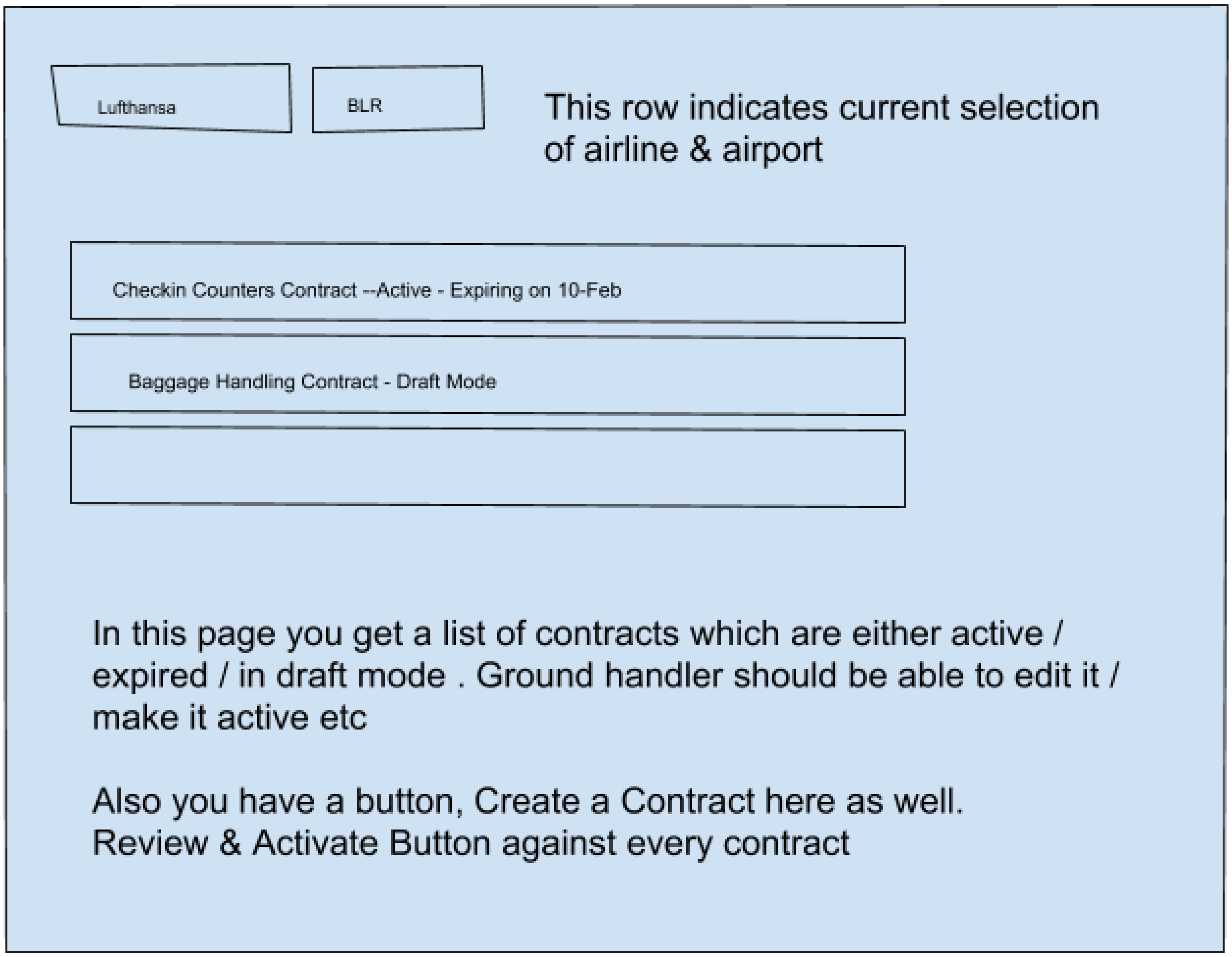
2) Contracts - Landing page



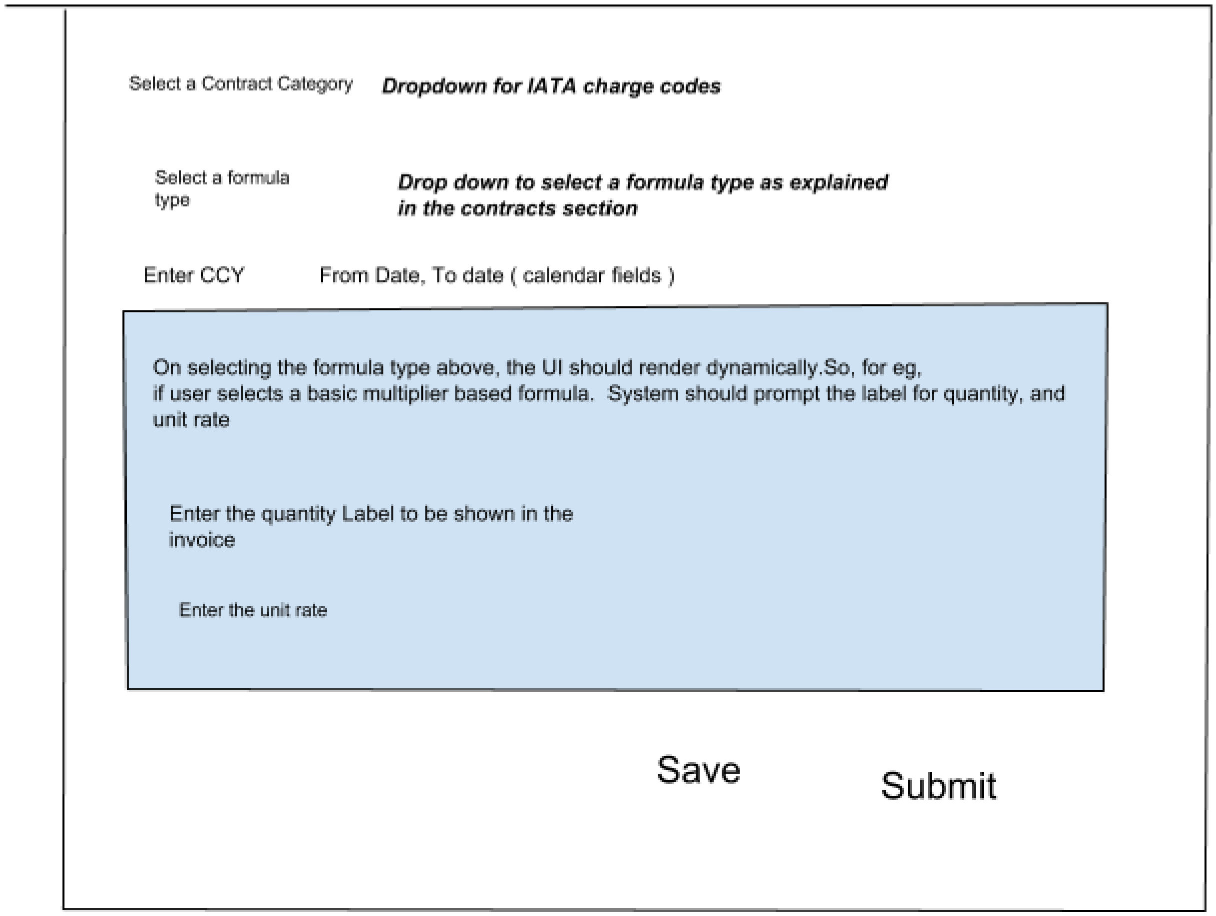
3) Contracts Airline View



4) Contracts List (on selecting airline & airport)



5) Contract Entry



Invoice Entry

Invoice Summary (Filters for
Draft , Airport, Airline